

EMPLOYMENT

Visiting Assistant Professor, University of Denver (DU), Denver, CO, USA	2025 – present
Predoctoral Fellow, University of Vermont (UVM), Burlington, VT, USA	2024 – 2025
Vermont Space Grant Consortium Graduate Fellowship	
Graduate Primary Instructor, UVM	2019 – 2021, 2024
Graduate Research Assistant, UVM	2022 – 2024
NASA EPSCoR Award	

EDUCATION

PhD in Mathematics, UVM	May 2025
Dissertation: <i>Odd Covers and Graph Saturation</i>	
Advisor: Puck Rombach	
MSc in Mathematical Sciences, UVM	May 2022
BA in Mathematics and English, UVM	Dec 2017

RESEARCH

Fields of interest

Combinatorics; Graph theory

Publications

Path odd-covers of graphs, with S. Borgwardt, E. Culver, B. Frederickson, P. Rombach, and Y. Yoo; *Discrete Mathematics*, 439(9):115078, 2026.

Trail Trap: A variant of Partizan Edge Geography, with M. Carr, A. Clifton, S. G. Hartke, V. Iršič Chenoweth, N. Sieger, and R. Whitman; *Discrete Applied Mathematics*, 379:79–95, 2026.

A lower bound on the saturation number and a strengthening for triangle-free graphs, with P. Rombach; *The Electronic Journal of Combinatorics*, 32(3), #P3.14, 2025.

Tropical geometry of Rado matroids, with R. Danner; *Discrete Mathematics Letters*, 14:27–30, 2024.

Node placement to maximize reliability of a communication network with application to satellite swarms, C.B., J. Bagrow, P. Rombach, and H. Ossareh; in *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Honolulu, Oahu, HI, USA; 3466–3473.

Toughness of recursively partitionable graphs, with B. Du Preez, K. E. Perry, and P. Rombach; *Theory and Applications of Graphs*, 10(2), Article 4, 2023.

Odd covers of graphs, with A. Clifton, E. Culver, J. Nie, J. O’Neill, P. Rombach, and M. Yin; *Journal of Graph Theory*, 104:420–439, 2023.

On the last new vertex visited by a random walk in a directed graph, with P. Horn and P. Rombach; *Discrete Mathematics Letters*, 11:96–98, 2023.

Subgraph complementation and minimum rank, with C. Purcell and P. Rombach; *The Electronic Journal of Combinatorics*, 29(1), #P1.38, 2022.

Accepted for publication

On cycles in multiset permutations, parking functions, and related structures, with F. Burghart, S. Wagner, and M. Yin; to appear in the proceedings of the 37th International Conference on Probabilistic, Combinatorial and Asymptotic Methods for the Analysis of Algorithms (AofA2026). (Preprint: *arXiv:2606.19500*, 2026.)

On odd covers of cliques and disjoint unions, with A. Clifton, E. Culver, P. Frankl, J. Nie, K. Ozeki, P. Rombach, and M. Yin; to appear in *Journal of Graph Theory*. (Preprint: *arXiv:2408.08598*, 2024.)

Submitted

On fixing and distinguishing numbers of trees, with P. Dankelmann, I. Harris, P. Horn, K. E. Perry, and E. Rivett-Carnac. (Preprint: *arXiv:2603.23820*, 2026.)

A Hierarchical Control Architecture for Optimal Guidance and Navigation of Satellite Formations, Y. Pedari, H. Basu, C.B., J. Bagrow, L.D. Espinosa, M.R. Almassalkhi, and H.R. Ossareh, 2026.

Vertex addition to a ball graph with application to reliability and area coverage in autonomous swarms, C.B., P. Rombach, J. Bagrow, and H. R. Ossareh. (Preprint: *arXiv:2506.19197*, 2025.)

Technical reports

One Algorithm to Rule Them All, with N. Brown, N. Chicoine, B. Emerick, A. Emmanuel, C. Hammond, K. Oluwasegun, O. Pomerenk, L. Schneegans, and H. Taboe. *Mathematics in Industry Reports* (submitted April 2025).

TEACHING*

Visiting Assistant Professor, DU

MATH 2060: Elements of Linear Algebra W 2026

MATH 1150: Foundations Seminar on the Mathematics of Voting and Counting W, S 2026

MATH 1953: Calculus III F 2025, S 2026

Graduate Primary Instructor, UVM

MATH 1212: Fundamentals of Calculus I F 2019, 2020, 2021

MATH 1077: Exploring Modern Mathematics S 2020, 2021, 2024

Instructor of Conversational English 2018 – 2019

Collèges Le Grand Beauregard and La Coutancière, La Chapelle sur Erdre, France

Teaching Assistant 2017, 2018

Johns Hopkins Center for Talented Youth, Bristol, RI

Course: Paradoxes and Infinities

Tutor

UVM Tutoring Center, Burlington, VT 2016 – 2018

Math Help Sessions, UVM 2019 – 2024

MENTORING

Graduate Mentor, Polymath Jr. REU 2024

*DU's Fall, Winter, and Spring quarters denoted F, W, and S, respectively; UVM's Fall and Spring semesters similarly.

Project group leader, Graduate Research Workshop in Combinatorics 2021, 2022

AWARDS

Vermont Space Grant Consortium Graduate Fellowship, UVM 2024 – 2025
 Nam Sang Kil Scholarship in Mathematics, UVM 2023
 Graduate Teaching Assistant of the Year, Lecture Instruction Category, UVM 2021
 John F. Kenney Award, Excellence in mathematics, UVM 2021
 AMS Joint Mathematics Meetings Graduate Student Travel Grant 2021, 2024
 AMS Graduate Student Sectional Travel Grant 2023
 UVM Graduate College Mini-Grant 2022, 2023
 Nominated to attend Mathematical Sciences Research Institute (MSRI), Berkeley, CA
 Random Graphs summer graduate school 2022
 The Phi Beta Kappa Society, Alpha of Vermont 2017

SERVICE

Math Outreach
 Lectured at Governor's Institutes of Vermont and MathCounts NW VT Chapter Competition 2023
 Prepared lessons on combinatorics for high school students (materials available on website)
 Graduate Student Council 2022 – 2024
 College of Engineering and Mathematical Sciences, UVM

ORGANIZING

DU Graduate Colloquium, Denver, CO (organizer) 2025 – 2026
 DU Analysis and Dynamics Seminar, Denver, CO (co-organizer) 2025 – 2026
 Joint Mathematics Meetings, Seattle, WA 2025
 AMS Special Session on Graphs, Hypergraphs, and Extremal Combinatorics (co-organizer)
 AMS Spring Central Sectional Meeting, Cincinnati, OH 2023
 Special Session on Extremal Graph Theory (co-organizer)
 UVM Combinatorics Seminar, Burlington, VT (co-organizer) 2023

WORKSHOPS

SAMSA Masamu Advanced Study Institute 2020, 2021, 2023, 2025
 15th Emléktábla Workshop – Extremal combinatorics group 2024
 SIAM Mathematical Problems in Industry Workshop 2024
 Graduate Research Workshop in Combinatorics 2021, 2022
 MSRI Random Graphs Summer Graduate School 2022

TALKS GIVEN

34th Cumberland Conference, Auburn University, Auburn, AL	May 2026
Masamu Virtual CRN Colloquium	Apr 2026
AMS Spring Eastern Sectional Meeting,* Boston College, Boston, MA	Mar 2026
DU Analysis and Dynamics Seminar,* Denver, CO	Oct 2025
AMS Spring Eastern Sectional Meeting,* Hartford, CT	Apr 2025
Iowa State University Discrete Math Seminar,* virtual	Feb 2025
Joint Mathematics Meetings,* Seattle, WA	Jan 2025
AMS Special Session on <i>Coloring Problems in Combinatorics</i>	
AMS Special Session on <i>Random Walks on Graphs and Related Parameters</i>	
Summit280, Budapest, Hungary	Jul 2024
NY Graph Theory Workshop,* CUNY Graduate Center, New York, NY	May 2024
NY Combinatorics Seminar,* CUNY Graduate Center, New York, NY	May 2024
AMS Spring Eastern Sectional Meeting,* Howard University, Washington, DC	Apr 2024
AMS Special Session on <i>Recent Trends in Graph Theory</i>	
Dartmouth-UVM Math Day, Dartmouth College, Hanover, NH	Feb 2024
Joint Mathematics Meetings,* San Francisco, CA	Jan 2024
AMS Special Session on <i>Extremal and Probabilistic Combinatorics</i>	
Southern Africa Mathematical Sciences Association (SAMSA) Annual Conference	Nov 2023
University of Pretoria, Pretoria, South Africa	
IEEE Int. Conf. on Systems, Man, and Cybernetics, Honolulu, HI	Oct 2023
AMS Fall Eastern Sectional Meeting,* University at Buffalo (SUNY), Buffalo, NY	Sep 2023
Governor's Institute of Vermont (outreach),* UVM, Burlington, VT	Jul 2023
Rutgers Graduate Combinatorics Seminar,* virtual	Apr 2023
54th Southeastern International Conference on Combinatorics, Graph Theory & Computing (SEICCGTC), Florida Atlantic University, Boca Raton, FL	Mar 2023
MathCounts Northwest VT Chapter Competition (outreach),* UVM, Burlington, VT	Feb 2023
Iowa State University Discrete Math Seminar,* virtual	Jan 2023
Joint Mathematics Meetings, Boston, MA	Jan 2023
AMS Special Session on <i>Research from the Graduate Research Workshop in Combinatorics (GRWC)</i>	
UVM Combinatorics Seminar,* UVM, Burlington, VT	Nov 2022
Dartmouth Combinatorics Seminar,* Dartmouth College, Hanover, NH	Nov 2022
Virginia Commonwealth University Discrete Math Seminar,* virtual	Oct 2022
SIAM DM22, Carnegie Mellon University, Pittsburgh, PA	Jun 2022
53rd SEICCGTC, virtual	Mar 2022
UMass Amherst Discrete Math Seminar,* Amherst, MA	Mar 2022
UVM Combinatorics Seminar,* UVM, Burlington, VT	Jan 2022

*Indicates an invited talk

AMS Fall Central Sectional Meeting, virtual	Oct 2021
UVM Combinatorics Seminar,* UVM, Burlington, VT	Sep 2021
SIAM DM21, virtual	Jul 2021
Northeast Combinatorics Network Discrete Math Day, virtual	Apr 2021
Graduate Students Combinatorics Conference, virtual	Apr 2021
63rd Midwestern Graph Theory Conference (MIGHTY), virtual	Apr 2021
52nd SEICCGTC, virtual	Mar 2021

RELEVANT SKILLS

Coding: Python, $\text{\LaTeX}$ and TikZ, HTML, SageMath

Languages: English (native), French (advanced)